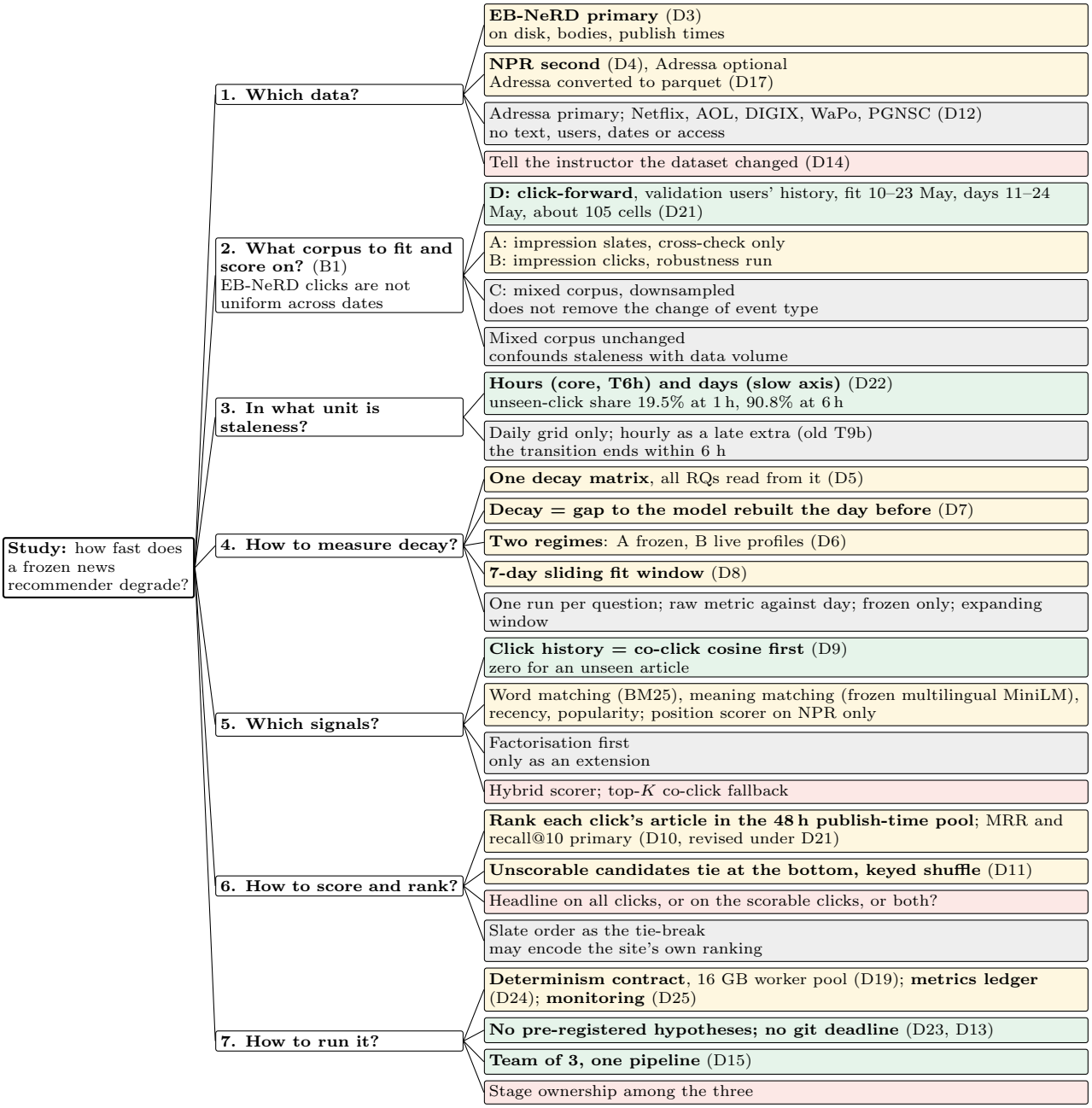
CS4.406 Term Project — Temporal Degradation of a News Recommender: Decision Tree So Far

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What this is. The study asks how fast a news recommender built from past clicks goes stale, which of three signals (word matching, meaning matching, click history) goes stale first, and how often a rebuild pays off. This note lays out the decisions taken so far as a tree: each node is a question, the branch taken is shaded, the alternatives rejected are grey with the reason, and open questions are red. **Shading:** decided proposed and acted on open rejected. No decay result exists yet; every number below is a measurement of the data, not of a model.

1 The tree



Read it as: the first three questions fix *what is measured on which data and in which unit*; the next three fix *how a model is scored*; the last fixes *how the work is run.* Everything above the open (red) nodes is decided or acted on; nothing red blocks the next stage.

2 Why the branches were taken [measured evidence]

Node	Evidence behind the branch (95% bootstrap intervals over fit days)
Corpus (B1)	EB-NeRD clicks per day fall from 9–13 M (4–17 May, both cohorts) to 1.6–2.0 M (26 May on) and change kind from history reads to impression clicks; a 7-day window ending 30 May holds about a fifth of the clicks of one ending 17 May. Only 0.4–0.7% of (user, article) pairs appear in both sources.
Unit (hours)	Share of clicks on articles published after the fit, 07:00 fit, 13 fit days, n = 3.4–4.6 M clicks per hour offset: 19.5 [14.6, 23.9]% at 1 h, 77.2 [72.7, 81.1]% at 3 h, 90.8 [88.5, 92.5]% at 6 h, 96.2 [95.8, 96.5]% at 24 h. The same curve on Adressa (80 fit days) reaches 48.5 [45.4, 51.5]% at 6 h and 87.8 [86.8, 88.7]% at 24 h; on NPR (85 fit days) 15.5 [11.3, 20.1]% at 6 h and 85.6 [80.6, 89.7]% at 24 h. Hours to 50%: 2, 7 and 13. The fit hour matters on EB-NeRD (6 h at a midnight fit, 2 h at 06:00) and barely on NPR.
Scoring	A frozen click-history model can score only articles seen in its window; at the smoke cell (fit 16 May, day 18 May, n = 19,939 clicks) 0.6% of the clicked articles were scorable. Seeded-shuffle MRR 0.0152 [0.0144, 0.0160]; recency 0.1773 [0.1739, 0.1805]. These are one cell, a pipeline check, not a decay result.
Cost	Click-history fit 18.6 s and 7.1 GB peak (planned 1.5 GB), so two workers at a time in the 16 GB pool; the other scorers fit in under 2 s.

3 Decision register

ID	State	Decision	Date
D1, D2, D12	decided	submitted proposal archived; Adressa copied by hand, not extracted; seven candidate datasets dropped	10-26
D3, D4	proposed, acted on	EB-NeRD primary; NPR second, Adressa optional	10-26
D5–D8	proposed, acted on	one decay matrix; two regimes; decay against the day-before rebuild; 7-day sliding window	10-26
D9	decided	click history = co-click cosine first	10-28
D10, D11	proposed, acted on	MRR and recall@10 primary under D21 (AUC and nDCG@5 secondary); keyed shuffle for ties	10-26
D13	decided	git deferred, no deadline attached	10-29
D14	open	tell the instructor about the dataset change	10-28
D15	decided	team of three, one pipeline	10-28
D16	superseded	optional hourly extension (by D22)	10-27
D17, D18	done	Adressa converted to parquet by streaming (3.11 GB, 113.6 M events); EB-NeRD copied and checksum-verified	10-27
D19	proposed, acted on	determinism contract; 16 GB pool; tested (1, 3 and 7 workers give identical sample files)	10-27
D20	done	two plans kept in `docs/plan/` and cross-checked	10-28
B1 / D21	decided	click-forward evaluation (option D), slate protocol as cross-check	10-29
D22	decided	hourly fast axis is core (T6h)	10-29
D23	decided	hypothesis practice removed (my addition, not a requirement)	10-29
D24, D25	decided	metrics ledger, serving proxies, planted-decay test, cleanup stage; monitoring through `monitor.py` (built, 10 tests)	10-29

4 What is built, what is open

Built and measured (`code/`, 30 tests passing): stores for the three datasets (EB-NeRD 113.4 M clicks; NPR 17.6 M, including all 1,402,576 impression clicks; Adressa 35.2 M), GPU embeddings for all three, the fixed sample of 20,000 clicks a day for 11–24 May, the model-free curves reproduced from the stores, and five scorers with a seeded shuffle on one smoke cell. **Not built:** the leakage and determinism gate (T5), the decay matrix (T6), the hourly stage (T6h), the analysis (T7), the rebuild-schedule replay (T8), the second dataset run (T9).

Open decisions (red nodes): (1) tell the instructor the dataset changed (D14, low stakes); (2) whether the headline metric counts unscorable clicks or reports the scorable clicks beside the share scorable (I would report both); (3) who owns which stage; (4) whether to build the hybrid scorer and the top-K co-click fallback, which were cut from the first pass. **Checks that remain:** an unexplained result in the smoke cell (a fresher frozen profile scored slightly lower for meaning and word matching, finding P41) and the EB-NeRD time zone, which the daily pattern suggests is UTC but which is not proven.

Rejected, with the reason kept: Adressa as primary (no impression lists, unfinished download at the time); a mixed or downsampled corpus (confounds staleness with data volume, or event type); a daily grid alone (the transition is over within hours on EB-NeRD); factorisation first (it would hide the zero-for-unseen mechanism under test); slate order as a tie-break (it may carry the site's own ranking).